A Dictionary for Matching Coefficients

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Abstract

The matching coefficients of an orthogonal-polynomial family — the coefficients approximating the weighted functions $W^{1/2}\widehat{P}_m$ by the polynomials $\widehat{P}_k$ — have been computed for ten families in a series of standalone papers. Each computation, taken alone, deflates to a classical mechanism. What does not deflate is the three-column language in which the computations become entries: a dictionary between analysis (generating-function kernels), combinatorics (transfer-matrix automata) and arithmetic (generator lattices). This paper records the dictionary, the four claims only it can state, and the one family — Wilson — on which it makes a falsifiable prediction.

1 The Dictionary

The Askey scheme contains almost no theorems and is among the most-used objects in the field. This paper is an object of the same kind.

analysis	**combinatorics**	**arithmetic**
GF kernel	automaton	generator lattice
ODE order of the kernel	number of states	lattice rank
Taylor coefficients u_n at base point	walk weights	all signs and generators
Cayley/Laguerre substitution	species chassis	universal, sign-free
kernel singularity type	—	obstruction level
D-module reducibility	—	algebraic vs. transcendental generators

Four sentences can be said in this language and, so far as we know, in no classical one.

1. *Rank classifies.* The Gaussian lattice has rank 1; Meixner–Pollaczek has rank 2 over Bessel (resp. Tricomi) values; continuous Hahn has rank 2 with one *algebraic* generator, the Jordan–Hölder filtration of its reducible D-module visible in the lattice; Wilson admits no finite rank at all. No classical reference makes this classification.

2. *The combinatorics is universal; the family chooses only u_n.* The defect–chain–matching chassis is identical across families; the weight enters through the prefactor exponent and the Taylor coefficients u_n of the kernel, which carry every sign. Proved for MP at $\lambda = \frac{1}{4}$ and, by changing two lines of the proof, at $\lambda = \frac{3}{4}$.

3. *Degeneration is evaluation, not limit.* The monic Jacobi matching entries are rational functions of the parameter α; the Chebyshev closed forms are their values at $\alpha = -\frac{1}{2}$, not an analytic limit.

4. *Obstructions form a hierarchy.* Entire weight $\to$ geometric decay and integrality over $\mathbb{Z}$; branch point at finite distance $\to$ power-law decay, lattice survives; Barnes factor $\Gamma(2ix)$ $\to$ no lattice.

2 The Chassis

Fix a weight W on $\mathbb{R}$ with orthonormal polynomials $\widehat{P}_n$ and set $\Phi_m = W^{1/2}\widehat{P}_m$ (for the Gamma-type weights, the analytic split $\Gamma(\lambda+ix)\widehat{P}_m$ replaces the square root). The *forward approximation* coefficients are $c_{mk} = \langle \Phi_m, \widehat{P}_k\rangle_W$; the *reverse expansion* coefficients are $\langle \widehat{P}_m, \widehat{P}_k\rangle_{W^{1/2}}$. Both are organised by a master generating function

$$\sum_{m,k} c_{mk}\, w^m u^k = \text{prefactor}(w,u)\; g(\zeta), \qquad \zeta = \varphi(w)\varphi(u),$$

with φ a Cayley or Laguerre substitution. The substitution factors as $\zeta = (1+\sigma)(1+\tau)$ in rescaled variables, and extracting coefficients produces one combinatorial chassis for every family: defect sets from the prefactor (double-factorial weights), chains from σ, τ (Lah numbers), and a Charlier-type partial matching of chains. The kernel g contributes only its Taylor coefficients at the base point $\zeta = 1$.

When g satisfies a linear ODE of order r with polynomial coefficients, differentiating the ODE at the base point yields an integer recurrence for the rescaled coefficients u_n, hence $u_n \in \mathbb{Z}A_1 \oplus \cdots \oplus \mathbb{Z}A_r$ for the r initial values A_i. This is the whole arithmetic mechanism. The worked instance is MP at $\lambda = \frac{1}{4}$, where $g = e^{\zeta/2}K_0(\zeta/2) = \sqrt{\pi}\,U(\frac{1}{2}, 1, \zeta)$ gives $u_{n+2} = -2n\,u_{n+1} + (4n+2)\,u_n$ with $u_0 = A$, $u_1 = A - B$ [6]; at $\lambda = \frac{3}{4}$ the same two steps give $u'_{n+2} = -2n\,u'_{n+1} + (4n+6)\,u'_n$ over the Tricomi generators [7]. Legendre's formula for the 2-adic valuation of $n!$ clears every denominator; the walk weights are positive integers and every sign of every table lives in u_n.

3 The Entries

One paragraph per family; proofs and tables are in the standalone papers.

family	GF kernel	order	lattice / generators	status
Hermite	exp(quadratic)	1	$\mathbb{Z}$; signed bipartite matchings	proved [2]
Laguerre	$2/(1+w+u-3wu)$	1	$\mathbb{Z}$; Delannoy numbers	proved [3]
Charlier	Charlier kernel	1	$\mathbb{Z}$; one-state automaton	proved [4]
Chebyshev–Jacobi	Wallis/Beta mass	—	quartic classes over $\Gamma(\frac{1}{4})^2/\sqrt{2\pi}$	proved [5]
MP $\lambda = \frac{1}{4}$, forward	$e^{\zeta/2}K_0(\zeta/2)$	2	$\mathbb{Z}A \oplus \mathbb{Z}B$, $A, B = \sqrt{2e}\,K_{0,1}(\frac{1}{2})$	proved [6]
MP $\lambda = \frac{1}{4}$, reverse	$e^{-1/\zeta}$	1	$(2\pi/e)\mathbb{Z}$	proved [6]
MP $\lambda = \frac{3}{4}$	$U(\frac{3}{2}, 1, \zeta)$	2	$\mathbb{Z}A' \oplus \mathbb{Z}B'$, $A', B' = U(\frac{3}{2},1,1), U(\frac{5}{2},2,1)$	proved [7]
unbalanced $\Gamma_{1/4}\Gamma_{3/4}$	first-order	1	single $A'' = e\,E_1(1)$	proved [8]
continuous Hahn	$\sqrt{\zeta}\,U(1,1,\cdot)$-type	2	$\mathbb{Q}\mathcal{A} \oplus \mathbb{Q}\mathcal{B}$, $\mathcal{A} = \sqrt{\pi}e^2E_1(2)$, $\mathcal{B} = \sqrt{\pi}$	empirical [9]
Wilson	Barnes $\Gamma(2ix)$	≥ 3?	none found to 10^{18}	open [10]

Three of the entries carry the dictionary's weight. The MP pair is the proved instance of claim 2. Continuous Hahn is the instance of the reducibility row, its kernel built on $e^z E_1(z)$, whose D-module is an extension of the trivial module by a Kummer module; the lattice sees the filtration through the algebraic generator $\sqrt{\pi}$, where the irreducible MP module has two transcendental generators. Chebyshev is the instance of claim 3, recovered from Jacobi by evaluating rational functions of α at $-\frac{1}{2}$ [5].

The Weil explicit formula provides a remark in the same language. In the MP $\lambda = \frac{1}{4}$ basis, the archimedean side compresses to the rank-2 lattice $\mathbb{Q}\sqrt{2\pi} \oplus \mathbb{Q}\sqrt{2\pi}(\gamma + \log 8\pi)$ (proved), while the prime side shows no $\mathbb{Q}$-compression among 16 entries (empirical) — the explicit formula's two halves are algebraically incommensurable in this basis [11].

4 The Obstruction Hierarchy

The decay of c_{0k} and the arithmetic degrade together, and the kernel's singularity type is the common cause.

1. *Entire weight* (Hermite, Laguerre, Charlier). Geometric decay; integrality over $\mathbb{Z}$; closed combinatorial formulas.

2. *Branch point at finite distance* (MP, Hahn; the branch point of $\Gamma(\lambda+ix)^2\Gamma(\lambda-ix)$-type kernels at $i/4$). Power-law decay of the coefficients (exponent $\approx \frac{3}{4}$ at $\lambda=\frac{1}{4}$); the lattice survives, with rank the ODE order.

3. *Barnes factor* $\Gamma(2ix)$ (Wilson). No lattice found; see §5.

The dual-pair construction repairs arithmetic, not analysis: replacing the orthogonal basis by the biorthogonal pair $\Gamma(\lambda \pm ix)\widehat{P}_m$ restores integrality at $\lambda=\frac{1}{4}$ without improving the L^2 truncation rate.

5 The Test

The dictionary predicts that lattice rank equals the ODE order of the GF kernel. Wilson is the falsifiable case.

Conjecture 5.1. *The generating-function kernel of the Wilson matching coefficients satisfies no linear ODE of order ≤ 2 with polynomial coefficients, and correspondingly the coefficients generate a $\mathbb{Q}$-vector space of unbounded dimension as $m,k\to\infty$.*

Evidence. The entries c_{mk}, $m,k\leq 5$, are $\mathbb{Q}$-combinations of 11 moments; PSLQ at 200 digits with coefficient bound 10^{18} finds no relation among the moments beyond the exact polynomial-product identities, so their $\mathbb{Q}$-rank is ≥ 16 counting both parity sectors — against MP and Hahn, which compress to 2 generators with coefficients below 10^8. What is missing is the analytic half: a computation of the D-module rank of the Wilson kernel itself. If that rank were finite and small, the dictionary would be wrong, and the failure would be as informative as the confirmations. If the language cannot make the Wilson question well-posed — the kernel row of the table is presently conjectural for Wilson — that too should be recorded, and this section is the record.

6 Status of the Claims

claim	status	source
Hermite, Laguerre, Charlier integrality	proved	[2, 3, 4]
Chebyshev–Jacobi classes; connection-Gram	proved	[5]
MP $\frac{1}{4}$ forward $\mathbb{Z}A\oplus\mathbb{Z}B$	proved	[6]
MP $\frac{1}{4}$ reverse $(2\pi/e)\mathbb{Z}$	proved	[6]
MP $\frac{3}{4}$ $\mathbb{Z}A'\oplus\mathbb{Z}B'$	proved	[7]
unbalanced, single generator	proved	[8]
Hahn rank-2 lattice	empirical (PSLQ)	[9]
Hahn generator $\sqrt{\pi}$ from reducibility	structural reading	[9]
rank = ODE order, in general	predicted	—
Wilson, no finite lattice	empirical (to 10^{18})	[10]
Weil archimedean rank 2	proved	[11]
Weil prime side rank ≥ 16	empirical	[11]

Empirical entries are PSLQ identifications at ≥ 40 digits, confirmed at doubled precision with coefficient bound $\geq 10^{10}$; no empirical claim is used in the proof of any proved claim.

References

[1] R. Langer and Claude Code, *Approximating Orthogonal Functions with Orthogonal Polynomials*, preprint, 2026.

[2] *The Hermite Matching Matrix*, preprint, 2026.

[3] *The Laguerre Matching Matrix*, preprint, 2026.

[4] *The Charlier Matching Matrix*, preprint, 2026.

[5] *The Jacobi Matching Matrix*, preprint, 2026.

[6] *The Dual Meixner–Pollaczek Matching Coefficients: the Calculation*, preprint, 2026.

[7] *The Dual Meixner–Pollaczek Matching Coefficients at* $\lambda = \frac{3}{4}$, preprint, 2026.

[8] *The Unbalanced Dual Pair*, preprint, 2026.

[9] *Continuous Hahn Matching Coefficients*, preprint, 2026.

[10] *Wilson Matching Coefficients*, preprint, 2026.

[11] *The Weil Form in the Meixner–Pollaczek Basis*, preprint, 2026.